

Introduction to Contract Theory and Auctions

Bayesian Games

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- The Buyer-Seller example from one of the previous lecture was an example of a **Bayesian Game**.
- In these games there is uncertainty over the payoff relevant (or possibly payoff irrelevant also) information in given pure strategy profiles.

Definition 2.1

A Bayesian game is a tuple $G = (N, \{A_i\}_{i \in N}, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N}, \{\mu_i\}_{i \in N}, \{u_i\}_{i \in N})$ where

- $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N}, \{\mu_i\}_{i \in N})$ is a belief space.
- A_i is the set of **actions** of player i . $A = \prod_{i=1}^n A_i$ is the set of **action profiles**.
- $u_i : A \times \Omega \rightarrow \mathbb{R}$ is players i 's (state-dependent) utility function.

In some applications: $\Omega = T = \prod_{i=1}^n T_i$, $\tau_i(t) = t_i \rightarrow u_i : A \times T_i$ (private values)
 $\Omega = T_0 \times T_1 \times \dots \times T_n$; $\tau_i(t_0, \underbrace{t_1, \dots, t_n}_t) = t_i$ $u_i : A \times T$ (interdependent values)

Definition 2.2

A **pure strategy** for player i is a function $s_i : T_i \rightarrow A_i$. A strategy of a player is a choice of action given her private signal realization. The **set of pure strategies for player i** is S_i and is the collection of all functions from T_i to A_i . The set of **pure strategy profiles** is $S = \prod_{i=1}^n S_i$.

$$|T_i| < \infty, |A_i| < \infty; |A_i|^{|T_i|}$$

Definition 2.3

A **behavioral strategy** for player i is a function $\sigma_i : T_i \rightarrow \Delta(A_i)$, that is, it is a choice of a probability distribution over actions given her private signal realization. The **set of behavioral strategies for player i** , $\hat{S}_i$, is the collection of all functions from T_i to $\Delta(A_i)$. The set of behavioral strategy profiles is $\hat{S} = \prod_{i=1}^n \hat{S}_i$.

Note $\Delta(S_i) \neq \hat{S}_i$ (Remark: in games of perfect recall, mixed & behavioral strategies can be shown to induce the same distribution over $\Delta(A)$)

Definition 2.4

Given a pure strategy profile $s = (s_1, \dots, s_n) \in S$, **player i 's ex-ante expected utility or payoff** is

$$E_{\mu_i, s}[u_i] = \int_{\Omega} u_i(s_1(\tau_1(\omega)), \dots, s_n(\tau_n(\omega)), \omega) d\mu_i(\omega)$$

Definition 2.5

Given a pure strategy profile $s = (s_1, \dots, s_n) \in S$, **player i 's interim expected utility or payoff** is

$$E_{\mu_i, s}[u_i | \tau_i = t_i] = \int_{\Omega} u_i(s_1(\tau_1(\omega)), \dots, s_i(t_i), \dots, s_n(\tau_n(\omega)), \omega) d\mu_i(\omega | \tau_i = t_i)$$

Definition 2.6

A **Bayes-Nash Equilibrium at the ex-ante stage** is a strategy profile $s^* = (s_1^*, \dots, s_n^*)$ such that no player can improve her ex-ante expected utility by changing unilaterally her strategy. That is, for all $i \in N$, all $s'_i \in S_i$, it holds that

$$E_{\mu_i, s^*}[u_i] \geq E_{\mu_i, (s'_i, s_{-i}^*)}[u_i]$$

Definition 2.7

A **Bayes-Nash Equilibrium at the interim stage** is a strategy profile s^* in which, for each player i and each type $t_i \in T_i$, it holds that

$$E_{\mu_i, s^*}[u_i | \tau_i = t_i] \geq E_{\mu_i, (s'_i, s_{-i}^*)}[u_i | \tau_i = t_i]$$

for all $s'_i \in S_i$.

Proposition 2.8

$BNE\text{-interim} \subseteq BNE\text{-ex-ante}$

Proof: Suppose s^* is BNE-interim

$$\begin{aligned} E_{\mu_i, s^*}[u_i] &= E_{\mu_i, s^*}[E_{\mu_i, s^*}[u_i | \tau_i]] \\ &\geq E_{\mu_i, s^*}[E_{\mu_i, s_i^*, s_{-i}^*}[u_i | \tau_i]] \\ &= E_{\mu_i, (s_i^*, s_{-i}^*)}[u_i] \end{aligned}$$

Remark 2.9

Conversely, if Ω is finite and there is a common prior $\mu = \mu_1 = \dots = \mu_n$, then if s is an ex-ante BNE, then there is an $\tilde{s}$ that is an interim-BNE, and such that the probability, under μ , that $s_i(t_i) \neq \tilde{s}_i(t_i)$ is zero.

Example: An opponent of unknown strength

The row player's type is known, but the column player can be strong, with probability α , or weak, with probability $1 - \alpha$. The payoff matrices are:

$u_{\text{row}}(F, F, S)$

		Column	
		F	Y
Row	F	-1, 1	1, 0
	Y	0, 1	0, 0

α

$u_{\text{row}}(F, F, w)$

		Column	
		F	Y
Row	F	1, -1	1, 0
	Y	0, 1	0, 0

$1 - \alpha$

$$\Omega = \{S, w\}, \quad \tau_1(S) = \tau_1(w) = t_1 ;$$

$$T_2 = \Omega ; \quad \mu(S) = \alpha, \quad \mu(w) = 1 - \alpha$$

$$\mu(S | \{S, w\} = t_1) = \frac{\alpha}{1} = \alpha$$

$$\tau_2(S) = S, \quad \tau_2(w) = w$$

$$\mu(S | \{S\}) = \frac{\alpha}{\alpha} = 1$$

$$\mu(w | \{w\}) = \frac{1 - \alpha}{1 - \alpha} = 1$$

$S_2^+(S) = F$ because when S_1 is dominant strategy to F .

If Row Fights she gets 1 if 2 weak

-1 if 2 strong

$$E_{\text{row}}(\text{Fighting}) = \alpha(-1) + (1-\alpha) \times 1 =$$

$$\text{Row ; Yield gets} = \alpha \times 0 + (1-\alpha) \times 0 = 0$$

$$\text{Row : F if } 1-2\alpha > 0 \Rightarrow \alpha < \frac{1}{2}$$

$$\text{Row Y if } 1-2\alpha < 0 \Rightarrow \alpha > \frac{1}{2}$$

$$S_1^* = \begin{cases} F & \text{if } \alpha < \frac{1}{2} \\ Y & \text{if } \alpha > \frac{1}{2} \end{cases}$$

The pure-strategy BNE = $\{(S_1^*, S_2^*(t_2))\}$

$$S_2^*(W) = \begin{cases} Y & \text{if } \alpha < \frac{1}{2} \\ F & \text{if } \alpha > \frac{1}{2} \end{cases}$$

Example: Classical Adverse Selection

A firm A is taking over another firm B . The true value of B is known to B but unknown to A ; it is only known to A to be uniformly distributed on $[0, 1]$. It is also known that B 's value will flourish under A 's ownership: it will rise to λx , where x is the pre-takeover value and $\lambda > 1$. This description is common knowledge.

$$\Omega = [0, 1], \quad \tau_A(x) = t,$$

$$N = \{A, B\}, \quad \tau_B(x) = x$$

$$T_1 = \{t, y\}, \quad T_2 = [0, 1]$$

$$b^+(x, y) = \begin{cases} 1 & \text{if } y \geq x \\ 0 & \text{if } y < x \end{cases}$$

$$\mu \sim \text{Uniform } [0, 1]$$

A strategy for A is a bid y (a number)

A strategy for B is Yes/NO decision
or a function $b(x, y) \in \{0, 1\}$

$$\text{Payoff of } A = \begin{cases} \lambda x - y & \text{if } b(x, y) = 1 \\ 0 & \text{if } b(x, y) = 0 \end{cases}$$

$$E[\text{payoff of } A] = \lambda E[X | y \geq X] - y = \lambda \frac{y}{2} - y = y \left(\frac{\lambda}{2} - 1 \right) > 0$$

$$Pr(X \leq y) = \int_0^y dx = y \quad \Leftrightarrow \lambda > 2$$

$$f(X | X \leq y) = \frac{1}{y} \text{ on } [0, y]$$

$$E[X | y \geq X] = \frac{1}{y} \int_0^y x dx = \frac{1}{y} \left. \frac{x^2}{2} \right|_0^y = \frac{y^2}{2y} = \frac{y}{2}$$

Example: Information can hurt!

In contrast with the single-agent case, more information can hurt both the informed and uninformed parties. Consider the following scenario: there are two states and two corresponding payoff matrices:

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

ω_1 : prob. 1/2

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

ω_2 : prob. 1/2

$N = \{R, C\}$, $\Omega = \{\omega_1, \omega_2\}$ $\tau_R(\omega) = \text{constant} = \tau_C(\omega)$ for all ω ; $\mu(\omega_1) = \mu(\omega_2) = \frac{1}{2}$
 each player has only one type $s_i(b_i) \leftrightarrow a_i$
 Trick: consider the "expected game"

forming payoff matrices.

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

$\omega_1 : \text{prob. } 1/2$

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

$\omega_2 : \text{prob. } 1/2$

		L	M	R
T	B	1, <u>2</u>	<u>1</u> , 1.5	<u>1</u> , 1.5
		<u>4</u> , <u>4</u>	0, 2.5	0, 2.5

The BNE is $\{B, L\}$

Example: Information can hurt! (continued)

Now suppose that player 2 is informed, while player 1 is uninformed, as before.

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

ω_1 : prob. 1/2

$T_1(\omega) = \text{constant for all } \omega$, $T_2(\omega) = \omega$

$s_2^*(a) = R$, $s_2^*(b) = M$

$s_1^* = T$

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

ω_2 : prob. 1/2

for all ω ; equally valid, $T_2 = \{a, b\}$

$T_2(\omega_1) = a$, $T_2(\omega_2) = b$

Example: Infection

Suppose that $T_1 = \{a, b\}$, $T_2 = \{a', b'\}$ Suppose the type maps are given by $\Omega = \{\omega_1, \omega_2, \omega_3\}$

$$\tau_1(\omega_1) = a, \tau_1(\omega_2) = \tau_1(\omega_3) = b, \quad \text{and} \quad \tau_2(\omega_1) = \tau_2(\omega_2) = a', \tau_2(\omega_3) = b'$$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

$$\mathcal{P}_1 = \{ \{\omega_1\}, \{\omega_2, \omega_3\} \}$$

$$\mathcal{P}_2 = \{ \{\omega_1, \omega_2\}, \{\omega_3\} \}$$

The "game" are events
 $\{\omega_1\}, \{\omega_2, \omega_3\}$

• At ω_1 , $K_1(\omega_1) = \{\omega_1\}$

$$K_2(\omega_1) = \emptyset \Rightarrow$$

$$K_1(K_2(\omega_1))^c = \Omega$$

$$(K_2(\omega_1))^c = \Omega$$

$$P_1 = \{ \{\omega_1\}, \{\omega_2, \omega_3\} \}$$

$$P_2 = \{ \{\omega_1, \omega_2\}, \{\omega_3\} \}$$

The "game" are events
 $\{\omega_1\}$, $\{\omega_2, \omega_3\}$

At ω_2

$$K_1 \{\omega_2, \omega_3\} = \{\omega_2, \omega_3\}$$

$$K_2 \{\omega_2, \omega_3\} = \{\omega_3\}$$

$$\rightarrow (K_2 \{\omega_2, \omega_3\})^c = \{\omega_1, \omega_2\}$$

$$\omega_2 \in (K_2 \{\omega_2, \omega_3\})^c$$

$$K_1 (K_2 \{\omega_2, \omega_3\})^c$$

$$= K_1 (\{\omega_1, \omega_2\}) = \{\omega_1\}$$

At ω_2 P1 doesn't know that P2 doesn't know the game.

$$\text{At } \omega_3 \quad \omega_3 \in K_1 \{\omega_2, \omega_3\} = \{\omega_2, \omega_3\} \Rightarrow \text{At } \omega_3 \text{ both know the game}$$

$$\omega_3 \in K_2 \{\omega_2, \omega_3\} = \{\omega_3\}$$

$$\omega_3 \notin K_1 K_2 \{\omega_2, \omega_3\} = K_1 \{\omega_3\} = \emptyset \Rightarrow \text{At } \omega_3 \text{ both know the game but P1 doesn't know if P2 knows it or not}$$

Strategies: $P_1: s_1(a), s_1(b)$ $\tau_1(\omega_1) = a, \tau_1(\omega_2) = \tau_1(\omega_3) = b$, and $\tau_2(\omega_1) = \tau_2(\omega_2) = a', \tau_2(\omega_3) = b'$

$P_2: s_2(a'), s_2(b')$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

$s_1(a) = D$ because is dominant for P_1 .

Now consider type a' of P_2 : $\Pr(\omega_1 | a') = \frac{3}{4} = \frac{9/13}{9/13 + 3/13}$

$$\Pr(\omega_2 | a') = \frac{1}{4}$$

$\rightarrow$ Playing L: payoff at most $\frac{3}{4} \cdot 0 + \frac{1}{4} \cdot 2 = \frac{1}{2}$
 Playing R: payoff is at least $\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 0 = \frac{3}{4}$

$\Rightarrow s_2(a') = R$

We look at P1 type b, posterior $(\frac{3}{4}, \frac{1}{4})$
 w_2, w_3

$s_2(a') = R$ so by playing:

$$\left. \begin{array}{l} U \text{ gets at most } \frac{3}{4} \times 0 + \frac{1}{4} \times 2 = \frac{1}{2} \\ D \text{ gets at least } \frac{3}{4} \times 1 + \frac{1}{4} \times 0 = \frac{3}{4} \end{array} \right\} s_1(b) = D$$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

Now consider $s_2(b')$ ($s_1(a) = D, s_2(a') = R, s_1(b) = D$)
 $\hookrightarrow s_2(b') = R$

$\hookrightarrow$ This is the infection:
 Outcome (U_L) is ruled out even
 when they both know the game.

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

Example: behavioral strategies

There are two players. Suppose that $T_1 = \{a, b\}$, $T_2 = \{c\}$, $\tau_1(t) = t_1$. The common prior is $\mu(a, c) = \mu(b, c) = 1/2$. The state games corresponding to each type vector are:

		Column	
		L	R
Row	T	1, 0	0, 2
	B	0, 3	1, 0

State game for $t = (a, c)$

		Column	
		L	R
Row	T	0, 2	1, 1
	B	1, 0	0, 2

State game for $t = (b, c)$

		Column		
		L q	R $1-q$	
Row	T	1, 0	0, 2	x
	B	0, 3	1, 0	$1-x$

State game for $t = (a, c)$

		Column		
		L q	R $1-q$	
Row	T	0, 2	1, 1	y
	B	1, 0	0, 2	$1-y$

State game for $t = (b, c)$

$$x = \Pr(\text{P1 of type a plays T})$$

$$y = \Pr(\text{P1 of type b plays T})$$

$$q = \Pr(\text{P2 plays L})$$

$$BR_1(q; t_1) = \begin{cases} x=1 & q > \frac{1}{2} \\ x \in [0, 1] & q = \frac{1}{2} \\ x=0 & q < \frac{1}{2} \end{cases}$$

$\downarrow$

$$(x_{BR}, y_{BR})$$

$$\begin{cases} y=1 & q < \frac{1}{2} \\ y \in [0, 1] & q = \frac{1}{2} \\ y=0 & q > \frac{1}{2} \end{cases}$$

$$q > \frac{1}{2}, t_1 = a$$

$$q = \frac{1}{2}, t_1 = a$$

$$q < \frac{1}{2}, t_1 = a$$

$$q < \frac{1}{2}, t_1 = b$$

$$q = \frac{1}{2}, t_1 = b$$

$$q > \frac{1}{2}, t_1 = b$$

$$BR_2(x, y) = \begin{cases} q=1 & 1 > 5x - 3y \\ q \in [0, 1] & 1 = 5x - 3y \\ q=0 & 1 < 5x - 3y \end{cases}$$

$$\text{if } 1 > 5x - 3y$$

$$1 = 5x - 3y$$

$$1 < 5x - 3y$$

$$L: \frac{1}{2} (x \cdot 0 + (1-x) \cdot 3) + \frac{1}{2} (y \cdot 2 + (1-y) \cdot 0)$$

$$R: \frac{1}{2} (2x) + \frac{1}{2} (y + 2(1-y))$$

		Column	
		L	R
Row	T	1, 0	0, 2
	B	0, 3	1, 0

State game for $t = (a, c)$

		Column	
		L	R
Row	T	0, 2	1, 1
	B	1, 0	0, 2

State game for $t = (b, c)$

$$BR_1(q; t_1) = \begin{cases} x=1 & q > \frac{1}{2}, t_1=a \\ x \in [0,1] & q = \frac{1}{2}, t_1=a \\ x=0 & q < \frac{1}{2}, t_1=a \end{cases}$$

$\downarrow$

$$(x_{eq}, y_{eq})$$

$$\begin{cases} y=1 & q < \frac{1}{2}, t_1=b \\ y \in [0,1] & q = \frac{1}{2}, t_1=b \\ y=0 & q > \frac{1}{2}, t_1=b \end{cases}$$

$$BR_2(x, y) = \begin{cases} q=1 & \text{if } 1 > 5x - 3y \\ q \in [0,1] & 1 = 5x - 3y \\ q=0 & 1 < 5x - 3y \end{cases}$$

if

$$L: \frac{1}{2}(x \cdot 0 + (1-x) \cdot 3) + \frac{1}{2}(y \cdot 2 + (1-y) \cdot 0)$$

$$R: \frac{1}{2}(2x) + \frac{1}{2}(y + 2(1-y))$$

~~A~~ If $q < \frac{1}{2} \Rightarrow x=0, y=1$
 $\Rightarrow$ violates (*) $\rightarrow$ can't be an eq.

If $q > \frac{1}{2} \Rightarrow x=1, y=0 \rightarrow$
 violates (*) $\rightarrow$ can't be an eq.

If $q = \frac{1}{2} \Rightarrow (x, y) \in [0,1]^2$ as
 long as they satisfy $1 = 5x - 3y$

The set of BNE (q, x, y) such that
 $\{q = \frac{1}{2}, x = \frac{1+3y}{5}, y \in [0,1]\}$

- If $q=1 \Rightarrow x=1, y=0 \Rightarrow q=0 \Rightarrow$ can't be an eq.
- If $q=0 \Rightarrow x=0, y=1 \Rightarrow q=1 \Rightarrow$ can't be an eq.

An eq. $q^* \in (0,1) \Rightarrow 1 = 5x - 3y$ (*). Which values (x, y) conform
 an equilibrium. To be an eq. (x, y) need to be a BR against
 q . ~~A~~

Example: Cournot's Duopoly with cost uncertainty

Two firms compete in selling a good: one firm does not know the other firm's cost function. Assume that both firms can produce the good at a constant unit cost. Assume also that they both know that firm 1's unit cost is c , but only firm 2 knows its own unit cost; firm 1 believes that firm 2's cost is c_L with probability θ and c_H with probability $1 - \theta$, where $0 < \theta < 1$ and $c_L < c_H$. Assume also that the inverse demand function is $P(Q)$ with $P(Q) > 0$ if $Q \in [0, \bar{Q})$ and 0 otherwise. Assume also that $P(Q)$ is differentiable.